ARTI 403 — Image Processing

Lab 3 — Image Manipulations using OpenCV Library

College of Computer Science and Information Technology — Department of Computer Engineering
Imam Abdulrahman Bin Faisal University — 2026 / 2027

Session topic	Image Manipulations using OpenCV Library
Session No.	3
Duration	150 minutes
Student name	Abdulrahman Saleh Alaamri
Student ID	2240002427

In [1]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

%matplotlib inline
```

Load the image

In [2]:
```
img = cv2.imread('../images/lena_gray_256.tif', cv2.IMREAD_GRAYSCALE)
h, w = img.shape

plt.imshow(img, cmap='gray')
plt.title('Original')
plt.axis('off')
plt.show()
```



Task 1: Geometric Transformations

1. Increase the size of the image

In [3]:
```
bigger = cv2.resize(img, (w * 2, h * 2), interpolation=cv2.INTER_LINEAR)

plt.imshow(bigger, cmap='gray')
plt.title(f'Resized to {bigger.shape[1]}x{bigger.shape[0]}')
plt.axis('off')
plt.show()

cv2.imwrite('output/resized.jpg', bigger)
```



Out[3]: True

2. Rotate the image by 120 degrees

In [4]:
```
center = (w // 2, h // 2)
M_rotate = cv2.getRotationMatrix2D(center, 120, 1.0)
rotated = cv2.warpAffine(img, M_rotate, (w, h))

plt.imshow(rotated, cmap='gray')
plt.title('Rotated 120 degrees')
plt.axis('off')
plt.show()

cv2.imwrite('output/rotated.jpg', rotated)
```



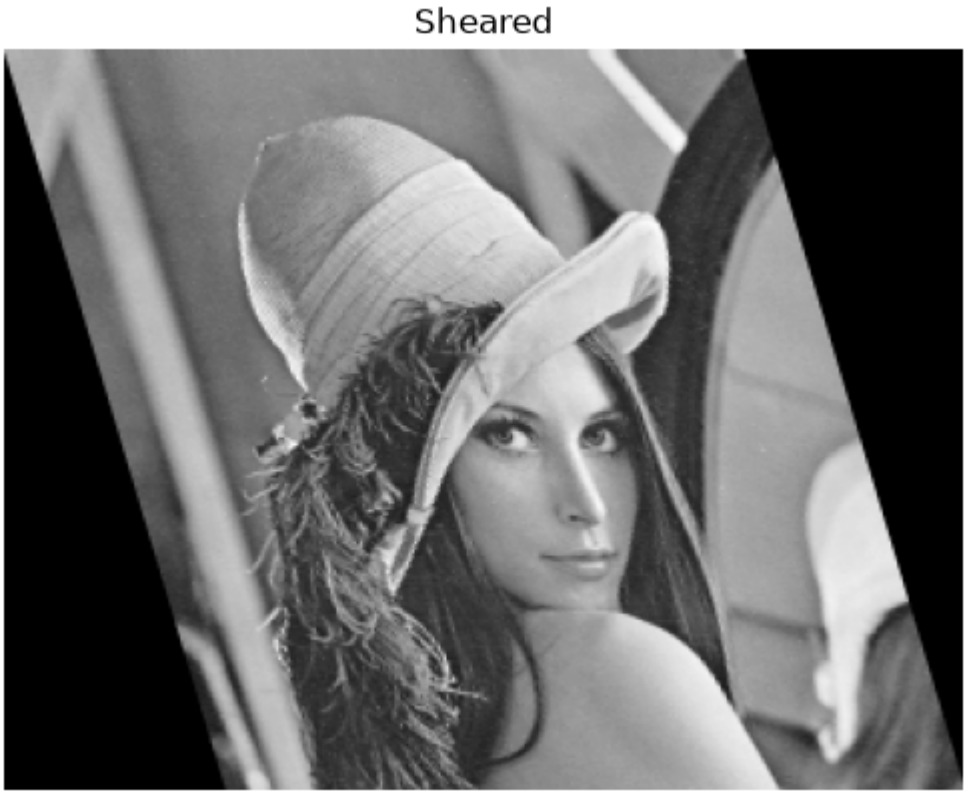
Out[4]: True

3. Shear operation

In [5]:
```
M_shear = np.float32([[1, 0.3, 0],
                      [0, 1, 0]])
sheared = cv2.warpAffine(img, M_shear, (int(w * 1.3), h))

plt.imshow(sheared, cmap='gray')
plt.title('Sheared')
plt.axis('off')
plt.show()

cv2.imwrite('output/sheared.jpg', sheared)
```



Out[5]: True

Task 2: Intensity Transformations

1. Negative image

In [6]:
```
negative = 255 - img

plt.imshow(negative, cmap='gray')
plt.title('Negative')
plt.axis('off')
plt.show()

cv2.imwrite('output/negative.jpg', negative)
```



Out[6]: True

2. Log transform

`c` is picked so the max output stays at 255.

In [7]:
```
c = 255 / np.log(1 + np.max(img))
log_img = c * np.log(1 + img.astype(np.float64))
log_img = log_img.astype(np.uint8)

plt.imshow(log_img, cmap='gray')
plt.title(f'Log transform (c={c:.2f})')
plt.axis('off')
plt.show()

cv2.imwrite('output/log.jpg', log_img)
```



Out[7]: True

3. Power law (gamma) transform

`gamma > 1` darkens the mid-tones and increases contrast.

In [8]:
```
gamma = 1.5
power_img = 255 * (img / 255) ** gamma
power_img = power_img.astype(np.uint8)

plt.imshow(power_img, cmap='gray')
plt.title(f'Power law (gamma={gamma})')
plt.axis('off')
plt.show()

cv2.imwrite('output/power_law.jpg', power_img)
```



Out[8]: True